Failure Analysis Report

FAIL-0345 — Synchronization Failure

1. Failure Summary

Failure ID	FAIL-0345
Failure Type	Synchronization Failure
Severity	Trivial
Component	Configuration Service
Application	APP-007 — InternetBrowser
Build Version	4.5.9
Timestamp	2024-12-26 20:21:10
Environment	ENV-003 — Hub Lab
OS Version	Tizen 7.0
Network	WiFi-6
Resolution	Resolved

2. Test Case Reference

Test Case ID	TC-1430
Title	Validate that health metric upload enforces access control when the network conn...
Execution ID	EXEC-01703
Test Run	RUN-008 — Regression Cycle 8 — Build 6.0.9
Requirement	REQ-0144 — Authentication — Remote Diagnostics

3. Expected vs Observed Behaviour

Expected Behaviour	The system should complete push notification delivery within the defined SLA.
Observed Behaviour	The system exhibited synchronization failure when the target device is running a legacy firmware ver...

4. Error Details

Error Message:

```
SyncError: conflict detected on entity 'ENTITY-5831'; last_modified timestamps diverge by 65s.
```

Stack Trace / Log Excerpt:

```
com.samsung.browser.ConfigurationService.execute(line 739)
```

Log Reference	LOG-99709

5. Executed Steps at Time of Failure

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1	Check server logs for event code 'EVT-3336'.	The UI transitions to the next screen within 3 seconds.
2	Launch the VoiceAssistant application on the test device.	Server returns HTTP 200 with a valid JSON payload.
3	Initiate voice command recognition by pressing the 'Pay' button.	Server returns HTTP 200 with a valid JSON payload.
4	Observe the status indicator for up to 29 seconds.	The fallback UI is displayed with a human-readable error message.
5	Verify that the UI displays 'Payment approved'.	The device reconnects automatically within 10 seconds.

6. Root Cause Analysis

Firmware update binary not validated against checksum before flashing; partial write leaves device in boot loop.

7. Linked Defect

Defect ID	BUG-0006
Title	Synchronization Failure in Configuration Service: SyncError: conflict detected o...
Severity	Trivial
Priority	Critical
Status	IN_PROGRESS
Root Cause	Firmware update binary not validated against checksum before flashing; partial w...
Resolution	—
Assigned To	ENG-072

8. Historical Similar Failures

HIST-0050	FAIL-0051	Synchronization Failure	3.1.9	2024-02-17	7	Wont Fix
HIST-0237	FAIL-0238	Synchronization Failure	6.4.3	2021-06-04	12	Resolved
HIST-0344	FAIL-0345	Synchronization Failure	3.0.9	2022-11-19	4	Open

9. Synthetic Data Notice

This document is entirely synthetic and intended for AI/RAG research only.

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